Demand Forecasting with Machine Learning in Retail: Case Study for Fawry Plus

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Abstract

This case study discusses using AI to help Fawry Plus predict future Yellow Card demand in its branches. The problem was that the traditional method of forecasting depended only on manual work, past averages, and personal judgment. This was not enough because demand changes from one branch to another and is affected by many variables like time, salary days, campaigns, seasons, and service problems.

The study will discuss how AI can help the company make better forecasts by using data from transactions, campaigns, and operational records. And the use of forecasting model to study old data, find patterns, and predict what may happen next. These results are then used in dashboards to help the company's departments and business functions to decide where more stock is needed, where more staff may be needed, and which branches need faster action.

Industry & Business Context

Introduction

The retail industry is a very dynamic industry that connects business and consumers through physical stores, digital platforms and hybrid models. It provides value for customers through the sale of physical products, services or a continuum of both.

The key challenge for business is managing demand accurately to balance product/service availability, inventory costs and supply chain. Forecasting demand is estimated by different quantitative and qualitative methods such as data, statistics and surveys.

Retail industry in Egypt is influenced by multiple factors like population growth, promotions, Regional demographics, Consumer behaviors and preferences, macroeconomics factors, seasonal events like Ramadan, public holidays, and even weather conditions. Analyzing multi demand variables using traditional methods makes forecasting inaccurate for the real world dynamic market. So the use of Artificial Intelligence and Machine Learning techniques will be very necessary for accurate forecasting and will enable the related data-driven decisions. And that's what will be explored in our case study.

Overview of the Selected Industry and Organization Type

The selected industry for this case study is retail fintech with a specific focus on Fawry Plus for Banking services company, which is the subsidiary and retail arm for Fawry for banking technology and electronic payment.

Fawry Plus operates a network of branches over 350 branches and provides wide range of services for its customers all over Egypt with scope of B2B Cash Collections/cash out, Wallets registrations, Remittance disbursements, KYC services for financial inclusion bank accounts, consumer finance, insurance, prepaid cards which includes also delivering these cards for customers in retail.

Demand in that context is not stable, and depends on many variables like branch location, time of day, day of the week, salary periods, public holidays, campaigns, seasonal events, and local customer behavior with huge volume of transactions history and operational data.

So Fawry Plus is highly suitable for the application of machine learning in demand forecasting. The scale of data, speed of delivering service and in high level decision related to staffing and inventory and variability of its demand factors create a strong case for AI demand forecasting solutions that can support faster, more accurate decision-making than traditional forecasting methods.

Key Business Challenges Faced Before AI Adoption

Before the AI adoption, Fawry Plus is facing several business challenges in forecasting branch demand accurately. The limitations of traditional methods that depends on historical transactions averages across services, branches on durations of month, quarter, or even the past year. Also they use subjective judgement of branches managers in requesting stock for prepaid cards and depend on planned direct requests from partners for huge amounts of cash.

But these factors are neglecting all other demand variability such as seasonality, Marketing campaigns and promotions, Regional differences, sudden or ad-hoc requests for cash, or even the CBE (Central Bank of Egypt) regulations and limitations in real time.

For example, CBE might change the cap on the maximum limited cash out transaction by beneficiary which might increase the demand on the amount of cash that should be available in branches.

Such dynamic variables are not considered in traditional methods as the effect of such changes is unknown and could not be forecasted using the current tools and resources.

So AI forecasting that considers seasonality trend, marketing campaigns, and real-time inputs either from the instant transactions that reflect current performance of each service in each branch over all branches with simulating different factors like CBE cash limitations and testing multiple scenarios is crucial for accurate planning either for staff utilization, inventory management and overall planning for company’s growth.

Market Conditions and Competitive Environment

Fawry Plus operates in a very digital transformational guided environment. Financial inclusion initiatives and the dependence for customers on suitable channels for payment and cash services, the expansion of fintech, wallets, banking agents all that increased the competition and made it a necessity for all fintech service providers to maintain a high level of service for availability, readiness, customer satisfaction, and efficient branches performance.

Problem Statement

Fawry Plus operates through a wide branch network that delivers financial, payment and banking services to its business or individual consumers across different locations in Egypt. Although the large amount of transactions and operations data, the company is unable to convert these data into accurate practical forecasts of branches demand which would support the daily operations, strategic decisions and overall growth.

The problem is more complex due to multidimensional variables for demand as it's not limited to one service or a stable pattern. Each branch experiences different levels of demand depending on its location, nearby population density, mix of operating services in this branch, customer profiles, payroll cycles, campaign effects, and local business activity. The decisions of operations supporting the branches also varies depending on such variables, as delivering prepaid cards stock in a branch in Cairo Area regarding the level of stock, the timing of it are different that delivering it to a branch in Aswan for example.

Another issue is that operational requirements in Fawry Plus are directly linked to demand accuracy. If future transaction volume is underestimated, and decision has been taken regarding the cash and stock based on that, a branch may suffer from cash shortages, insufficient card stock, long waiting times or staff understaffing. If demand is overestimated, the company may lock unnecessary liquidity in branches, over-allocate employees or creating issues in stock distribution and workforce planning.

So forecasting inaccuracies will lead to Inventory imbalance and inefficient staffing and poor customer experience and not just analytical problems, that will have negative consequences financially and operationally.

Literature review

Related studies show a clear shift from traditional forecasting methods toward machine learning and big data approaches because retail demand is influenced by many interacting variables. Karthikeyan and Vijay examined AI-powered demand forecasting from a retail sustainability perspective. Their study explained how the predictive analytics with machine learning can reduce inventory stocking imbalance and inefficient resource allocation. They highlighted techniques such as LSTM networks, random forests, gradient boosting, reinforcement learning, and hyperlocal demand prediction using NLP-based models. They also discussed practical results such as improved inventory efficiency, lower carbon impact and sustainability. However this study highlighted on the limitations in this forecasting process including data silos, infrastructure

gaps, algorithmic bias, and the difficulty of integrating AI systems into fragmented retail environments.

Yadav proposed a retail demand forecasting framework that combines machine learning with big data inputs. The research focused on integrating transactional data with social media activity, weather data, and calendar-based events to improve responsiveness to changing customer behavior. The main algorithms emphasized were Random Forest and XGBoost, supported by a cloud-based architecture and a React dashboard for visualization. The study concluded traditional methods such as ARIMA and linear regression are limited because they do not handle external variables well and can't adapt quickly to real-time demand shifts. Although the strong accuracy improvements, they identified recurring limitations such as computational complexity, scalability issues, data imbalance, weak cross-channel integration, and the need for stronger preprocessing.

The RELEX retail forecasting guide adds a practical industry view. It explained that machine learning improves demand forecasting by modeling recurring patterns, promotions, price elasticity, local events, and weather effects at detailed store or channel level. Its main emphasis is not one single algorithm, but the ability of machine learning systems to process large-scale multi-factor data and continuously improve forecasts. At the same time, the guide notes important practical limits, for specific products with little history, AI often guesses poorly because it lacks enough examples. This creates overfitting and treats randomness as a strict rule. To fix this, we need human oversight to ensure the "black box" stays logical and transparent.

And all these studies support the main idea of this case study: demand forecasting performs better and more accurately when it moves from static historical estimation to machine learning models that combine operational and real-time data. This is highly relevant to Fawry Plus where demand changes by branch, service type, timing, campaigns, and external conditions.

Data Analytics Component

Data Sources

The data analytics component in this case study is very important for converting the data collected from all sources across the company into practical actionable insights. This stage is used for organizing the data, detecting patterns, measure the relationships and correlations between different demand variables for accurate forecasting

The data used is structured, historical, and real-time operational data. The most important source is **SuperFawry** the platform used for registering customer's data and **POS** (point of sale) machines that records the payment transactions, both act as the core transactional source for branch activity. They provide detailed records of service performance across Fawry Plus branches and reflects the actual operational demand generated by customers. This includes transaction-level activity for services such as cash in/out, bill payments, prepaid cards, wallet-related services, remittances, consumer finance, insurance and all branch services. These records are the main structured dataset for forecasting because they capture what happened, where it happened, and when it happened.

Another key source is **Microsoft Dynamics Customer Insights.** This is the ERP module used to record the marketing campaigns data which is the source of marketing context in the model. Campaign schedules and promotional timing help explain why service demand may rise or fall beyond normal historical patterns. This is important because forecasting at Fawry Plus cannot rely only on past transactions. It must consider the impact of planned campaigns and monitor its impact on branches and service performance.

The last source in our model is **operational data**, and the source of this data is an integration with Data Center servers for the information of failures and outages. These variables reflect branch reality and explain abnormal transaction drops or disruptions that should not be treated as normal demand behavior. Including them improves forecasting quality because it separates true demand shifts from technical or operational incidents which will be considered as outliers.

Data Types

Regarding the dataset used in our model, it is built from **structured, historical, and real-time operational data**, which together form the analytical foundation of the forecasting model. **Structured data** refers to data stored in defined fields and columns inside operational databases, such as transaction numbers, branch codes, dates, service identifiers and campaign flags. This is the main form of data used in the model because machine learning algorithms require consistent and well-organized inputs.

Historical data is used to train the model on past branch behavior and recurring demand patterns such as salary periods, holidays, campaigns, and seasonal events. It helps the model learn how demand behaved previously and detect repeated patterns across branches and services.

Real-time operational data is used to reflect the current state of branch activity, including live transactions, outages, and service disruptions, so the forecasting process does not depend only on old averages but can adapt to actual operating conditions.

Before modeling, the data must be cleansed and standardized. The cleansing process includes removing duplicate records, validating branch codes and service identifiers, converting date and time fields into consistent formats, handling missing values, excluding cancelled or invalid transactions and converting event flags into numeric binary values. This preprocessing stage is essential because forecasting quality depends directly on data quality. As the model will only produce reliable forecasts if the dataset is accurate and complete.

The following tables summarize the main datasets, their sources, their role as model input or output, and the exact fields used in the database with the data type of each field:

Field Name	Likely Data Type	Description	Source system
transaction_no	Text / Varchar	Unique transaction reference number	SuperFawry / POS

mobile_number	Text / Varchar	Customer mobile number linked to the transaction	SuperFawry
branch_code	Text / Varchar	Unique identifier of the branch	SuperFawry / POS
branch_name	Text / Varchar	Name of the branch where the transaction occurred	SuperFawry / POS
area	Text / Varchar	Geographic area or governorate of the branch	SuperFawry / branch master data
date	Date	Transaction date	SuperFawry / POS
time	Time / Text	Transaction time	SuperFawry / POS
gender	Char / Text	Customer gender	SuperFawry
shift	Text / Varchar	Shift during which the transaction was executed	POS / branch operations
acquisition_channel	Text / Varchar	Source through which the customer was acquired	SuperFawry / CRM / marketing records
interest_yawmy	Integer	Binary indicator showing interest in Yawmy service	SuperFawry
is_anomaly	Integer	Binary flag indicating abnormal operational behavior	Data center / operational monitoring
anomaly_type	Text / Varchar	Type of anomaly such as outage or failure	Data center / operational monitoring
is_campaign_day	Integer	Binary flag indicating whether the date falls within a campaign period	MS Dynamics
is_post_eid	Integer	Binary flag indicating post-Eid period	Derived from business calendar
is_back_to_school	Integer	Binary flag indicating back-to-school season	Derived from business calendar
is_black_friday	Integer	Binary flag indicating Black Friday period	MS Dynamics

is_valentine	Integer	Binary flag indicating Valentine period	Derived from business calendar
is_salary_period	Integer	Binary flag indicating salary cycle period	Derived from business calendar / payroll cycle assumptions
branch_code	Text / Varchar	Branch identifier used to join transaction and master data	Branch master data
branch_name	Text / Varchar	Official branch name	Branch master data
area	Text / Varchar	Branch area or governorate	Branch master data
date	Date	Aggregated date for total daily demand	Derived from transactions
value	Real / Float	Total aggregated daily demand value	Derived from transactions
branch_code	Text / Varchar	Branch identifier for branch-level daily demand	Derived from transactions
date	Date	Date of branch-level aggregation	Derived from transactions
value	Real / Float	Daily aggregated demand value by branch	Derived from transactions
area	Text / Varchar	Area for area-level daily aggregation	Derived from transactions
date	Date	Date of area-level aggregation	Derived from transactions
value	Real / Float	Daily aggregated demand value by area	Derived from transactions
acquisition_channel	Text / Varchar	Acquisition channel used for channel-level aggregation	Derived from transactions / CRM
date	Date	Date of channel-level aggregation	Derived from transactions
value	Real / Float	Daily aggregated demand value by channel	Derived from transactions

start_time	Datetime	Start time of outage or operational incident	Data center / server monitoring
end_time	Datetime	End time of outage or operational incident	Data center / server monitoring
duration_minutes	Integer	Duration of the incident in minutes	Data center / server monitoring
severity	Text / Varchar	Severity level of the outage or incident	Data center / server monitoring

Table 1 - Input Data Fields Used in the Fawry Plus Demand Forecasting Model

Dataset.csv

Attached Also the dataset

AI Solution Overview

AI Technology Used

The AI technology used in this case study is a hybrid machine learning forecasting solution built around two main algorithms: **Prophet** and **XGBoost**. The reason for selecting a hybrid approach is that Fawry Plus demand is influenced by both recurring time-based patterns and more complicated relationships between factors.

A single forecasting model can capture part of this behavior, but combining two complementary models improves accuracy. In the project architecture, the forecasting firstly transforms raw branch transaction record into an analytical time series, then applies a baseline forecasting model, and finally adds a corrective learning layer before passing the results into operational and managerial decision support.

Prophet is the first forecasting stage. Prophet was introduced by Facebook's data science team as a scalable forecasting procedure for business time series with strong seasonality, holidays, and trend changes. Technically, Prophet is based on an additive modeling philosophy in which the observed series is decomposed into trend, seasonal components, and event effects. In business applications, its main advantage is that it can model recurring weekly and yearly patterns while also including specific events such as campaigns and holidays with relatively high interpretability. In this project, Prophet is configured with growth**,** weekly and yearly seasonality for data starting from Jan 2025 till End of Mar 2026, and all related variables.

The second algorithm is **XGBoost**, which stands for Extreme Gradient Boosting. Historically, XGBoost was formalized by Chen and Guestrin as a scalable tree-boosting system, building on

earlier gradient boosting research by Friedman. Its strength lies in learning complex nonlinear patterns, interactions between variables, and residual structures that simpler forecasting models may leave unexplained. In practical terms, once Prophet generates the baseline forecast, XGBoost is used in this project as a residual correction model. That means it studies the remaining error and learns how to correct it using additional features. This is especially useful when demand shifts are influenced by nonlinear operational factors rather than pure seasonality.

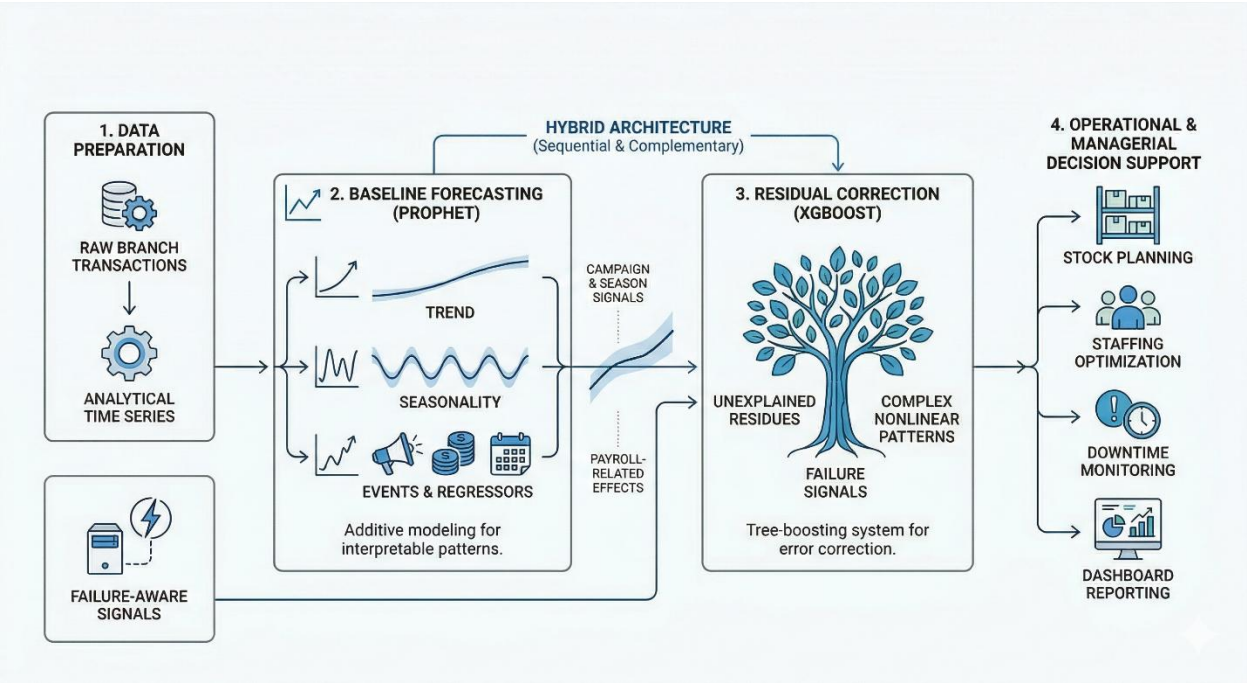

Figure 1 - The Hybrid Machine Learning Forecasting Solution for Fawry Plus

Role of Robotic Process Automation (RPA) in the Case Study

In general, **Robotic Process Automation (RPA)** refers to the use of software bots/scripts to automate repetitive tasks that cannot be performed manually by employees. RPA does not replace machine learning instead its role is to automate routine processes such as data extraction, cleansing, data transfer.

The first role of RPA in this case is **data collection and consolidation**. Fawry Plus demand forecasting depends on data coming from multiple data sources mentioned before, collecting this data from different systems, exporting it, cleaning it, and preparing it for forecasting would consume time and introduce human error. RPA can automate this sequence by extracting updated data from source systems at scheduled time.

The second role is **data preprocessing**. Data must be checked, standardized, and prepared before model. These tasks (validating data structure, checking missing values, formats standardization, fixing incomplete records) are done through RPA.

Implementation Methodology

SDLC is the most implementation approach or framework used in software worldwide, however CPMAI has a more suitable background for this case than a software-only SDLC approach. It was developed as a specialized methodology for managing AI and machine learning projects, and PMI later formalized it in its CPMAI v7 framework as a vendor-agnostic, data-centric, AI-specific, and iterative methodology. PMI then launched the PMI-CPMAI certification on September 30, 2025, replacing CPMAI v7 and reinforcing it as a dedicated lifecycle for AI initiatives rather than a traditional software delivery model. This evolution is important in this case because the Fawry Plus solution depends on business understanding, data readiness, model iteration, validation, deployment, and monitoring, which are central characteristics of AI projects.

In this case study, **CPMAI** is the most suitable methodology because the solution is not only a software development project. It is primarily an **AI and analytics implementation project** where the core value comes from transforming raw operational data into an intelligent forecasting system that supports real business decisions. Traditional SDLC models are useful when the main challenge is designing, coding, testing, and deploying a software application with stable and well-defined requirements. However, this case is more complex because success does not depend only on whether the system works, but on whether the data is prepared correctly, whether the model learns demand patterns, whether forecast accuracy is accepted, and if the outputs improve staffing, inventory, and branch planning. That is exactly the type of situation CPMAI is designed for.

The main justification for CPMAI is that the project follows an **analytics lifecycle**, not just an application lifecycle. The case begins with a business problem: Fawry Plus cannot forecast branch demand accurately using manual estimation and historical averages alone. From there, the project moves into understanding demand drivers, identifying useful data, preparing structured inputs, engineering forecasting features, selecting models, validating results, deploying outputs, and monitoring business impact. These steps go far beyond a normal software build sequence. In a traditional SDLC approach, the focus would usually be on gathering requirements, designing screens, building APIs, testing functionality, and launching the system. That would be useful for the dashboard, database, and workflow components, but not enough to manage the iterative and experimental nature of machine learning.

Another reason CPMAI is more appropriate is that **machine learning projects are highly dependent on data readiness and model iteration**. In this case, the team must explore historical demand behavior, study seasonality, identify campaign effects, engineer anomaly and failure features, and evaluate how well Prophet and XGBoost perform. These activities require repeated refinement, not a one-time fixed sequence. The project files clearly show feature engineering, forecasting layers, residual correction, evaluation metrics, and continuous refresh logic. This means the project depends on cycles of testing, learning, and improving. CPMAI fits this well because it is designed to move from business understanding to data understanding, then to data preparation, model building, evaluation, and operational deployment in a structured but iterative way.

If **traditional SDLC** were applied to this case, it would still have value, but only **partially**. SDLC would be relevant for the technical parts such as designing the application architecture,

building the database, creating dashboards, implementing APIs, managing user access, and testing the interface and system integration. In other words, SDLC can support the **software side** of the project. It can define how the platform is designed, coded, tested, deployed, and maintained. That part is complete and important, especially because the project includes dashboards, query layers, and operational interfaces.

The first CPMAI stage is **business understanding**. In this case, the business problem was clearly defined as the inability of Fawry Plus to forecast branch demand accurately using traditional methods. The company operates in a dynamic environment where demand changes according to campaigns, salary cycles, seasonal events, branch locations, and operational disruptions. Which means that the business objective was not only to build a technical forecasting model, but to improve branch readiness, reduce stock imbalance, support staffing decisions, and strengthen management visibility. This stage is critical because the forecasting solution must be designed for real operational pain points.

The second stage is **data understanding**. At this stage, the development team studies what data exists, how reliable it is, how often it changes, and how it reflects real branch behavior. In this case, the data is not treated as one simple dataset. This stage ensures that the forecasting problem is translated into measurable variables that the model can actually learn from.

The third stage is **data preparation**. This is one of the most important stages in the case because machine learning accuracy depends heavily on how the data is prepared. The project sources show that raw branch data is transformed into analytical layers. Here the transaction records are cleaned, structured, standardized, and converted into model-ready inputs. Missing dates are handled, seasonal and campaign features are added, and operational incidents are turned into usable forecasting signs.

The fourth stage is **model planning and model building**. In this case, the selected solution is a hybrid model using Prophet and XGBoost. Which was explained in previous sections.

The fifth stage is **evaluation**. Here, the model is tested to determine whether it solves the business problem well enough. In the project, forecast results are compared using measures such as MAE, RMSE, and MAPE and the outputs are visualized through analytical dashboards. Evaluation is important because a technically working model is not enough if it does not improve branch planning or management decisions. The model must prove that it is more useful than manual forecasting and historical averages.

The sixth stage is **deployment and operationalization**. This is where the project becomes a real business solution deployed on production and.

Finally, CPMAI also have **monitoring and continuous improvement phase**. New campaigns, new regulations, branch growth, and operational changes can affect performance over time. And so the solution must be refreshed, monitored, and recalibrated always.

Analytical Techniques (Applied)

In this case study, we use four main types of analytics.

Descriptive analytics means looking at old data to understand what already happened. It helps us see branch activity, daily transactions, busy areas, and service demand.

Diagnostic analytics means trying to understand why something happened. It helps us know why demand went up or down, such as campaigns, salary days, holidays, or system problems.

Predictive analytics means using old data to guess what may happen next. This is the main part of the project. The AI model studies past demand and predicts future demand.

Prescriptive analytics means deciding what action to take after the prediction. It helps the company know where to send more stock, where more staff are needed, and which branches need quick attention.

Key Performance Indicators (KPIs) and Metrics

The KPIs in this case help us know if the AI system is doing a good job. Some KPIs show how close the forecast is to the real demand, so we can see if the prediction is correct or not. Other KPIs show branch activity, such as daily demand, branch workload, and which branches are busier than others. There are also stock KPIs that help the company know the stock level, when to reorder, and which branches may run out of stock soon. Staffing KPIs show if a branch has enough employees, if more staff are needed, or if staff are not being used well. Finally, stability KPIs show if there are system problems, such as downtime, failures, or unusual activity. All of these KPIs help Fawry Plus understand performance better and make faster and smarter decisions.

To make the results easy to understand and use, the system includes three main dashboards: **operational**, **analytical**, and **managerial**. Each dashboard is made for a different level of decision-making. Together, they help the company move from raw data to clear actions.

The **operational dashboard** is used for daily work and quick action n. Its main goal is to show what needs attention now or in the next few days. It includes important cards such as **Stock Alerts**, **Utilization %**, **Downtime**, and **High Risk Branches**. It also shows lists and simple charts that help the team know which branches need stock first, where staff should be added or rebalanced, and where service is unstable. This dashboard is useful for branch operations teams because it focuses on urgent issues and short-term action.

The **analytical dashboard** is used to understand demand and forecast behavior over recent weeks or months. It includes measures such as **Transactions**, **Customers**, **Anomaly %**, and **Yawmy %**. It also contains visual charts like the **Daily Trend** line chart, the **Actual vs Forecast** chart, and the **Channel Share** chart. These views help users understand the overall demand direction, see how close the forecast is to the real result, and know which channel contributes

more. This dashboard is useful for analysts and business teams because it explains what is happening in the data.

The **managerial dashboard** is made for higher-level decisions. It gives a monthly or summary view and focuses on actions management should take. It includes KPIs such as **Need Refill**, **Need More Staff**, **Suggested Inbound**, and **Average Days to Stockout**. It also shows priority lists such as **Top Actions**, **Staff Watchlist**, and short summary notes. This helps management decide where to send stock, where extra staff are needed, and which branches should come first in planning.

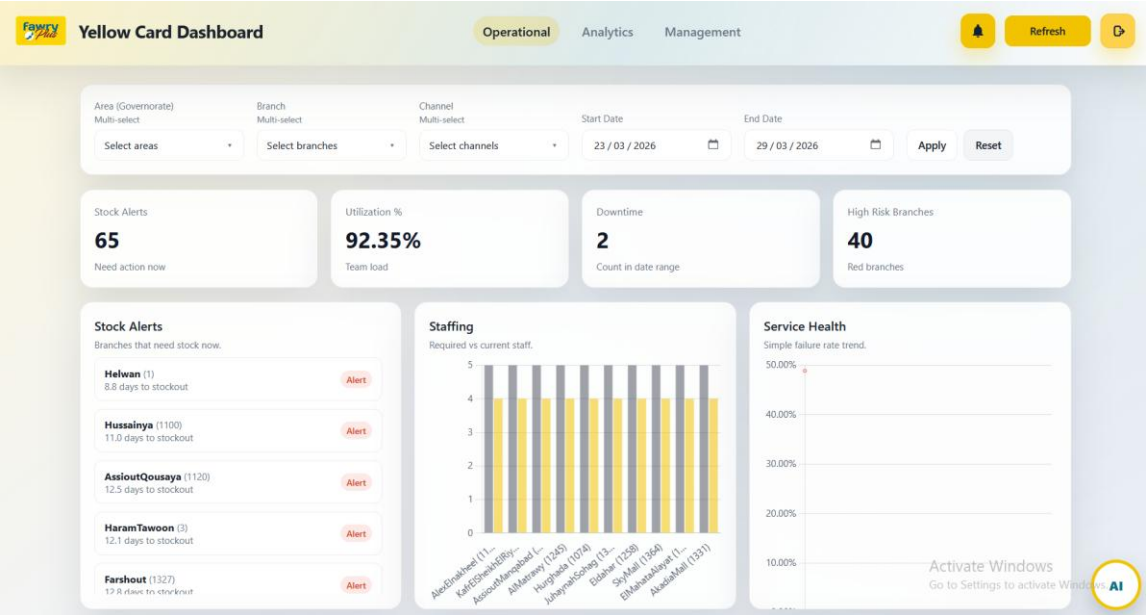

Figure 2 - Operational Dashboard

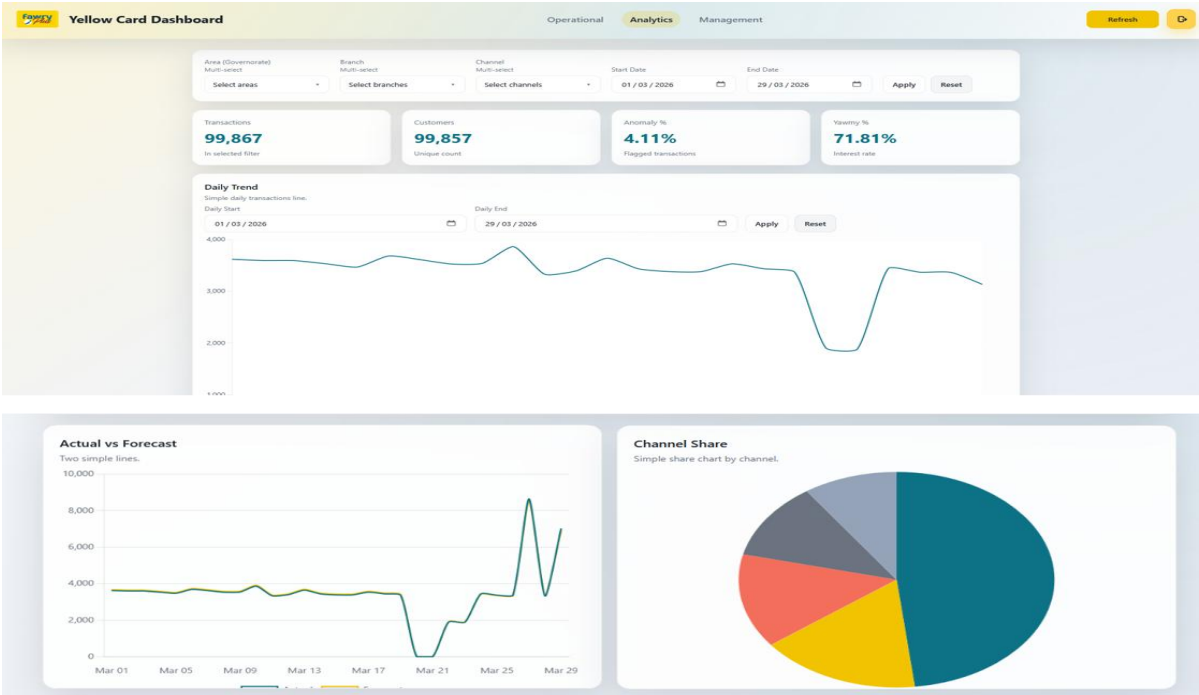

Figure 3 - Analytics Dashboard

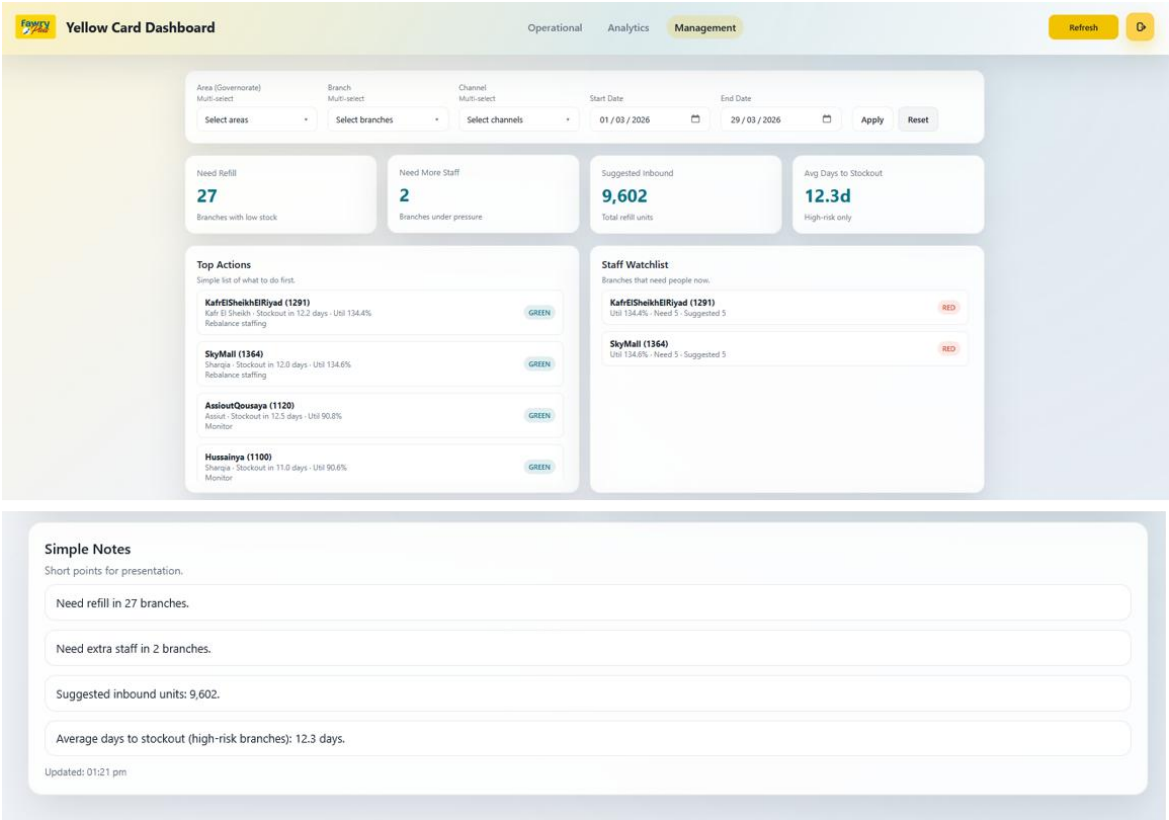

Figure 4 - Management Dashboard

Digital Transformation

In the Fawry Plus case, the difference before and after digital transformation is very clear. Before the transformation, branch demand planning depends more on manual work, old reports, and personal judgment. Teams looked at past branch activity and tried to estimate future demand using simple averages or experience. This made planning slower and less accurate, especially when demand changed because of salary periods, campaigns, seasonal events, branch location, or service problems. In many cases, decisions were taken after the problem already appeared, such as cash shortage, low card stock, long waiting time, or weak staff allocation.

After digital transformation and adopting AI and machine learning techniques, Fawry Plus moves to a data-driven way of working. Data from transactions, campaign activity, and operational incidents is brought together and used in one forecasting flow. Instead of only looking through historical transactions, the company can now look forward through AI forecasts. which helps the business expect changes earlier and prepare before the problem happens.

The daily work also changes. Before, planning was done in separate steps and updated from time to time. After the transformation, the process becomes more dynamic because dashboards and forecasting outputs are refreshed regularly. This means branch support teams can respond faster to changes in stock needs, staffing pressure, and service instability.

Decision-making improves too. Before, two managers might take different decisions based on their own judgment. After transformation, decisions become more consistent because they are based on shared dashboards, forecast results, and clear KPIs. Management can see which branches need refill, which branches need more staff, and which branches are high risk.

From the customer side, the service becomes more stable. When Fawry Plus can plan better for cash, prepaid cards, and staff, customers face fewer shortages, fewer delays, and a better branch experience.

So, in this case, digital transformation means moving from manual and reactive branch planning to faster, smarter, and more proactive planning based on data and AI.

ERP Integration & Role

ERP integration is very important in this case because it connects the AI forecasting model with the real business work inside Fawry Plus. The ERP side works like the main system that brings data from different business activities together and helps turn forecast results into real actions. This means the AI model does not work alone. It takes data from business systems, analyzes it, then sends useful outputs back to support planning and execution.

In this case, the main data comes from sources such as **SuperFawry**, **POS transactions**, **Microsoft Dynamics campaigns**, and **operational incident data** like failures and outages. These data sources are pulled into the ingestion layer through pipelines and automated processes. After that, the data moves to the processing layer, where it is cleaned, checked, and organized.

At this stage, the system prepares daily demand views by **branch**, **area**, and **channel**. This step is very important because the model needs clean and structured data before forecasting starts.

After data processing, the AI modeling layer starts working. In this project, the forecasting tool uses a **hybrid model** made of **Prophet** and **XGBoost**. Prophet helps with trend and seasonality, while XGBoost helps capture more difficult patterns and improve accuracy. The model then produces forecasts, metrics, and insights. These results are stored in a structured way so they can be used later in dashboards and decision-making.

The next step is the visualization layer. Here, the outputs appear in the **operational dashboard**, **analytical dashboard**, and **managerial dashboard**. This helps different teams understand the situation from their own point of view. For example, operations teams can see stock alerts and downtime, analysts can see forecast versus actual trends, and management can see branch priorities, staffing needs, and suggested inbound stock.

ERP integration becomes even more useful in the execution stage. Forecast results are not only displayed; they are also used to support real actions in the business. These actions may include inventory planning, workforce allocation, branch prioritization, finance follow-up, and marketing response. This creates a connected loop where data is collected, analyzed, turned into decisions, and then applied in the business system. After that, actual results and human feedback return again to the model through feedback loops, so the system can learn and improve over time.

The main benefits of this integration are better coordination, faster decisions, and more useful forecasting. Still, there are also challenges such as data quality, system compatibility, cost, security, and the need to keep the model updated. Even with these challenges, ERP and AI together give the company a much stronger way to manage demand than using separate reports or manual planning only.

The full solution architecture flow will be something like the following, and the attached figure can be used to show this clearly.

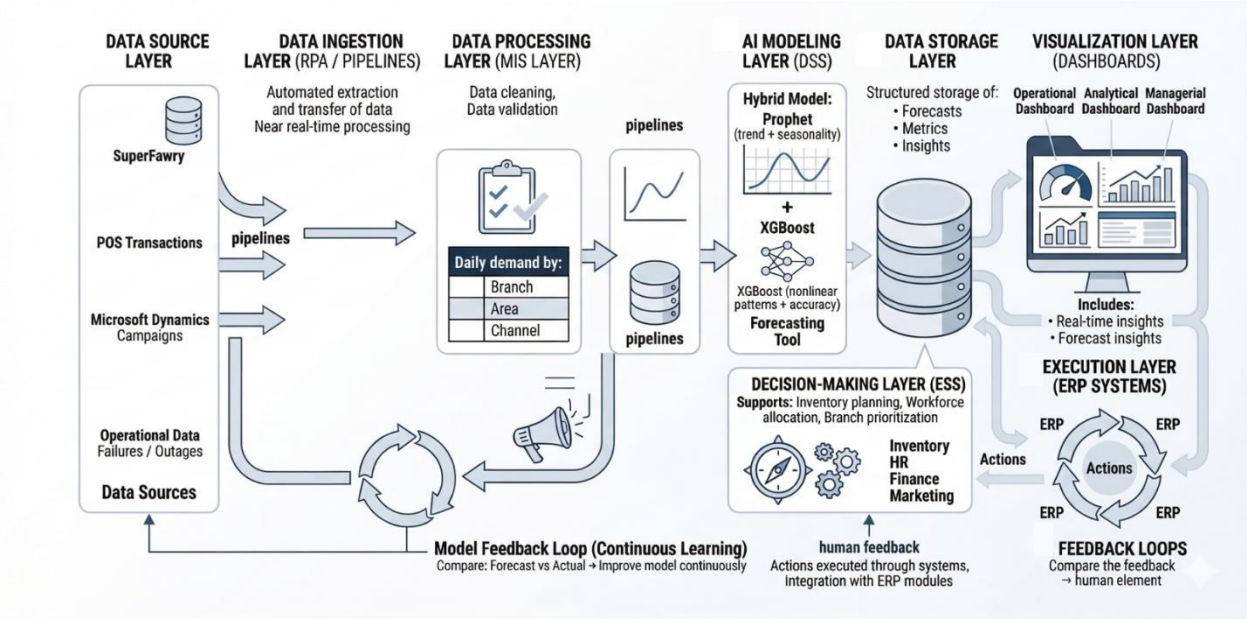

Figure 5- Full System Architecture Flow

Business Impact & Value Creation

AI-driven demand forecasting can bring strong business value to this case. First, it can improve financial results. When Fawry Plus knows expected demand earlier, it can prepare better for busy periods, reduce missed sales, and use promotions in a smarter way. This helps increase revenue and makes spending on marketing and operations more useful.

It also improves operational efficiency. Before AI, planning depended more on manual work and old reports. After using forecasting, the company can plan stock, staffing, and branch readiness in a better way. This reduces delays, lowers errors, and helps teams respond faster when demand changes.

Another benefit is strategic advantage. Fawry Plus can make better decisions than before because it is using future demand signals, not only past data. This makes the company more flexible, more responsive, and better prepared for changes in the market.

Finally, the solution supports long-term growth. As the company grows and handles more data, the forecasting system can continue to support more branches and more business needs. This makes the solution useful not only now, but also in the future.

Risks, Ethical Issues & Limitations

The first risk is **data privacy and security**. The system uses branch transactions, customer-related records, campaign data, and operational incident data. Even if the model does not use full

personal details, this data is still sensitive. If the data is not protected well, it may be seen or used by the wrong people. Because of this, Fawry Plus must control who can access the data, protect it while moving between systems, and store it in a secure way.

The second risk is **bias in the AI model**. The model learns from past branch activity. If old data has weak patterns, missing information, or unusual branch behavior, the model may learn the wrong thing. For example, if some branches had more campaigns than others in the past, the forecast may keep giving them more attention. This can affect stock planning, staffing, and branch priorities. To reduce this, the model should be checked often and updated with newer data.

There are also **regulatory and compliance issues**. Since the system works with customer and transaction data, Fawry Plus must follow privacy laws and internal company rules. The company must know what data is collected, why it is used, and who is allowed to use it. It should also keep records of access and usage.

Another challenge is **technology adoption risk**. Some employees may trust manual planning more than AI at the beginning. Others may not understand how the forecasting system works. Also, the company needs people who can monitor the model, review the results, and handle system updates. Without training and support, the system may not be used well.

There are also **limitations in AI and ERP systems**. The forecast is only as good as the data. If branch data is missing, late, or wrong, the forecast will also be weak. AI may also struggle with rare sudden events. On the ERP side, integration can be difficult, and some systems are not flexible enough for fast AI updates.

In this case, **ISO 27001** is very relevant. The most useful controls are **access control**, so only the right people can see sensitive data; **encryption and secure transfer**, so data is protected while moving between systems; **operations security**, so data processing stays accurate and safe; **secure system development and integration**, so the forecasting system is built in a safe way; and **incident management**, so Fawry Plus can react quickly if a security problem happens.

Strategic Recommendations for Improvement

To make this solution better in Fawry Plus, the company should keep improving the data and the model. One good step is to add more useful data, such as branch area details, stronger campaign information, and more live operational signals. Better data will help the forecast become more accurate and more useful for branch planning. The company should also keep checking the model results and retrain the model from time to time, because demand patterns can change.

Another important recommendation is to use the forecast more deeply in daily decisions. The forecast should not stay only in dashboards. It should help more in stock planning, staffing decisions, branch prioritization, and management follow-up. Employees should also be trained to understand the forecast results and use them correctly. This will help the company get more value from the system.

In the future, Fawry Plus can expand this project in many ways. The company can use more advanced AI models, include more real-time data, and improve the speed of response to sudden demand changes. It can also connect forecasting with dynamic planning, smarter alerts, and more automatic actions inside the system. Later, the same idea can support other services, not only Yellow Card. In this way, the project can grow from a forecasting tool into a wider AI-based decision support system for the business.

Conclusion

This case study showed how AI can help Fawry Plus improve demand forecasting for Yellow Card across its branches. The main problem was that old planning methods depended too much on manual work, past averages, and personal judgment. These methods were not enough because branch demand changes with time, location, campaigns, salary periods, and even service problems.

The study showed that AI gives Fawry Plus a better way to understand demand and prepare early. By using branch transactions, campaign data, and operational signals together, the company can build forecasts that are more accurate and more useful. This helps the business make better decisions about stock, staffing, branch readiness, and management priorities.

The case also showed that data analytics is very important. It helps turn raw branch data into clear insights, dashboards, KPIs, and forecasting outputs that support real decisions. Instead of only looking at what happened before, Fawry Plus can now look ahead and act before problems appear.

Another important point is that the solution is not only a model. It is part of a full business system that includes AI forecasting, ERP integration, dashboards, and decision support. Using CPMAI also made the project more suitable because it fits AI and data projects better than a normal software-only approach.

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